A group of students were given water from two reservoirs and asked to determine water hardness using flame atomic absorbance (AA) spectroscopy. Water hardness can be found by relating the concentration of calcium and magnesium ions to the concentration of carbonate species in the sample using Equation 1.

$$[CO_3^{2-}] = 2.5[Ca^{2+}] + 4.1[Mg^{2+}]$$

Equation 1

To correlate the concentration of metal ions to absorbance, the students made standard solutions of MgO (MW = 40.31 g/mol) and CaO (MW = 56.08 g/mol) by diluting small portions of 0.001 M MgO and 0.01 M CaO stock solutions with 2% HNO_3 into 100 mL volumetric flasks. The calibration curves generated from the standard solutions are shown in Figures 1 and 2.

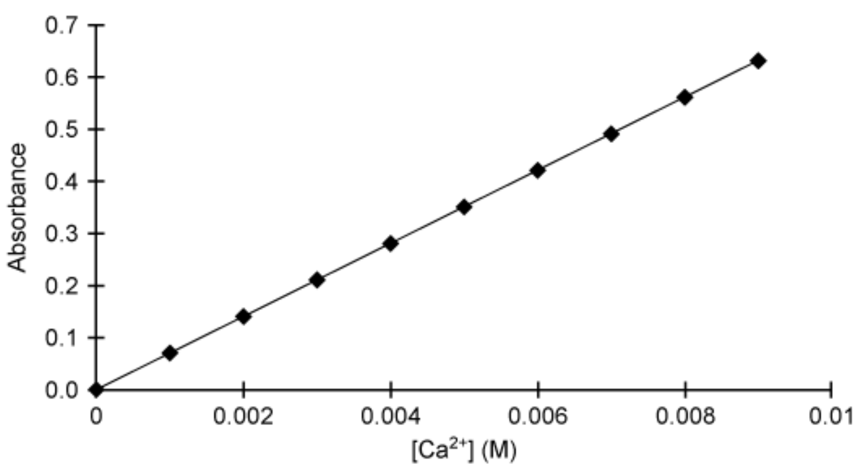


Figure 1 Calibration curve from the absorption of the Ca^{2+} solution standards

Ca^{2+} solution standards

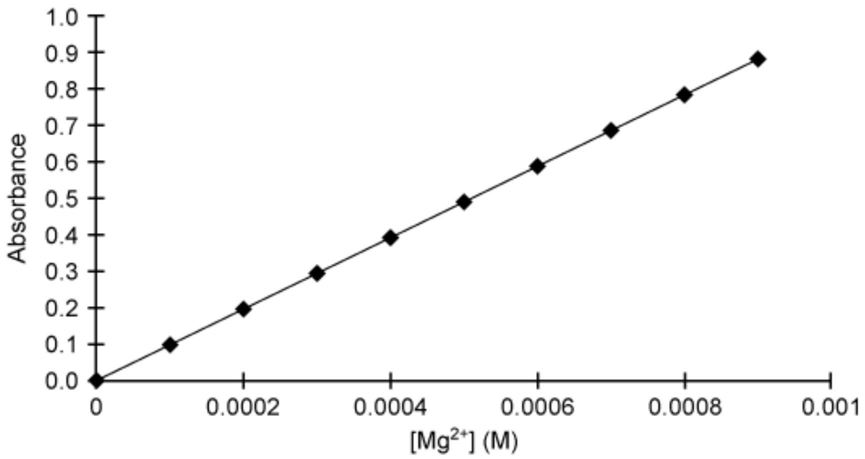


Figure 2 Calibration curve from the absorption of the Mg^{2+} solution standards

Lanthanum(III) chloride (MM = 245.25) can be used to eliminate chemical interference in a sample by competitively binding to PO_4^{3-}, releasing calcium to be atomized in the flame for detection.

The students prepared unknown Samples 1 and 2 by mixing 25 mL of reservoir water, 16 mL of 2% HNO_3, and 9 mL of 0.1 M $LaCl_3$ for a total sample volume of 50 mL each. The samples were each tested three times using flame AA spectroscopy, yielding the results reported in Table 1.

Table 1 AA Spectroscopy Results for Reservoir Water Samples

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

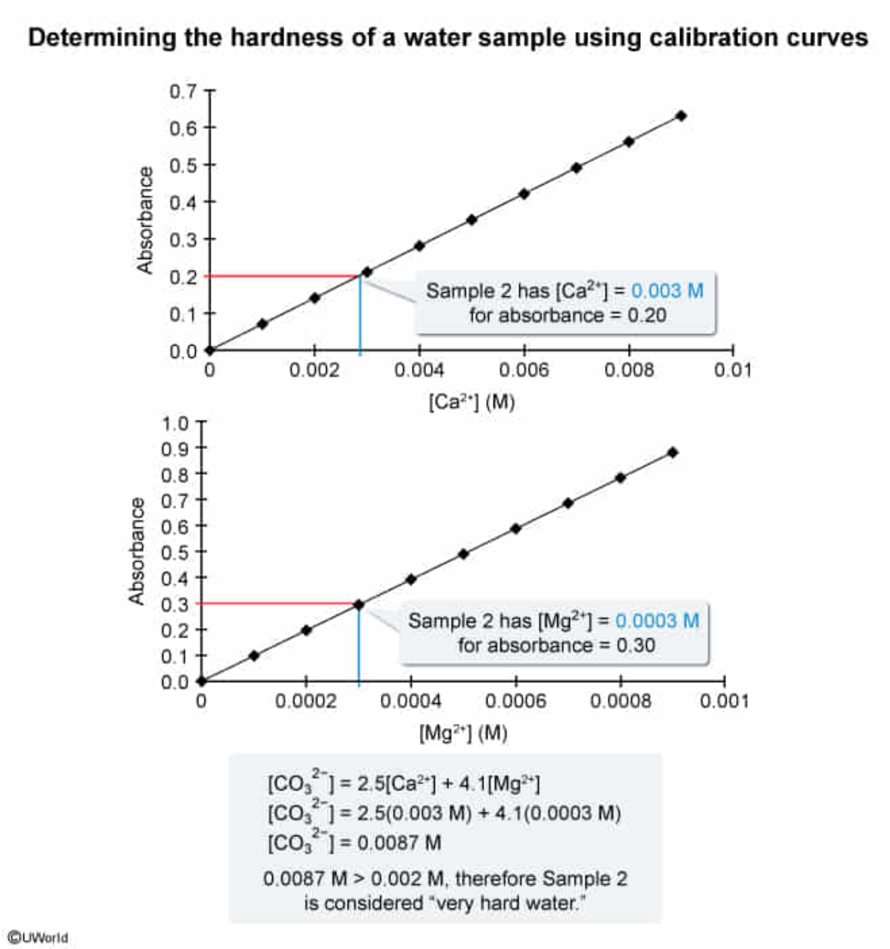
According to the passage, the "hardness" of a water source is based on the concentration of CO_3^{2-} species.

Water is considered to be "very hard" if $[CO_3^{2-}]$ is at least 0.002 M. After preparation, is the Sample 2 mixture considered "very hard water"?

- A. No; $[CO_3^{2-}]$ = 0.0012 M [8%]
- B. No; $[CO_3^{2-}]$ = 0.0017 M [19%]
- C. Yes; $[CO_3^{2-}]$ = 0.0030 M [22%]
- ✔ D. Yes; $[CO_3^{2-}]$ = 0.0087 M [50%]

Correct
50% answered correctly

Explanation:



Calibration curves are used to find a general relationship between the **concentration** of a compound that is being studied (**analyte**) and the instrumental detection. This is accomplished by testing standard solutions containing a known amount of the analyte. The resulting calibration curve can then be used in measurements of samples containing unknown quantities of the same analyte.

According to the passage, water hardness is determined using Equation 1:

$$[CO_3^{2-}] = 2.5[Ca^{2+}] + 4.1[Mg^{2+}]$$

where $[Ca^{2+}]$ and $[Mg^{2+}]$ are the molar concentrations of Ca^{2+} and Mg^{2+}, respectively. Using the absorbances in Table 1 and the calibration curves in Figures 1 and 2, $[Ca^{2+}]$ in Sample 2 is 0.003 M and $[Mg^{2+}]$ is 0.0003 M. Substituting these concentrations into Equation 1 gives:

$$[CO_3^{2-}] = (2.5)(0.003\ M) + (4.1)(0.0003\ M)$$

$$[CO_3^{2-}] = 0.0075\ M + 0.0012\ M = 0.0087\ M$$

Because 0.0087 is greater than 0.002 M, Sample 2 is considered "very hard water."

(Choices A and B) Although CO_3^{2-} concentrations of 0.0012 M or 0.0017 M indicate that the water should not be considered "very hard water," these concentrations are incorrect. A concentration of 0.0012 M is obtained if $[Mg^{2+}]$ alone were considered in the calculation, and 0.0017 M is obtained if an average of $[Ca^{2+}]$ and $[Mg^{2+}]$ were taken.

(Choice C) A concentration of 0.0030 M indicates the water should be considered "very hard water," but this value of $[CO_3^{2-}]$ is incorrect because it considers only $[Ca^{2+}]$. As defined by the passage, water hardness takes into account both $[Ca^{2+}]$ *and* $[Mg^{2+}]$ sources of carbonate in Equation 1 to determine $[CO_3^{2-}]$.

Educational objective:

The concentration of an analyte in an unknown sample can be determined by comparing it to a calibration curve prepared from samples of known analyte concentration. The determined concentration may be used in empirical relationships to assess sample characteristics.

Time spent: 0 sec

Subject:
General Chemistry

QID:401557

System:
Atomic Theory and Chemical Composition